

# CENG212-Devre Teorisi ve Uygulamaları

## H2CD2-Devre Elemanları

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# CHAPTER 2 *Circuit Elements*

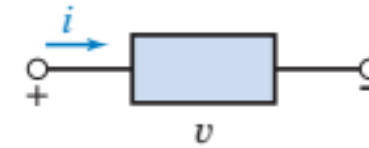
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## 2.2 Engineering and Linear Models



A **linear element** satisfies the properties of both superposition and homogeneity.

Then we may state the two properties required as follows.

*Superposition:*

$$i_1 \rightarrow v_1$$

$$i_2 \rightarrow v_2$$

then

$$i_1 + i_2 \rightarrow v_1 + v_2$$

*Homogeneity:*

$$i \rightarrow v$$

then

$$ki \rightarrow kv$$

A device that does not satisfy either the superposition or the homogeneity principle is said to be nonlinear.

### EXAMPLE 2.2-1 A Linear Device

Consider the element represented by the relationship between current and voltage as

$$v = Ri$$

Determine whether this device is linear.

### EXAMPLE 2.2-2 A Nonlinear Device

Now let us consider an element represented by the relationship between current and voltage:

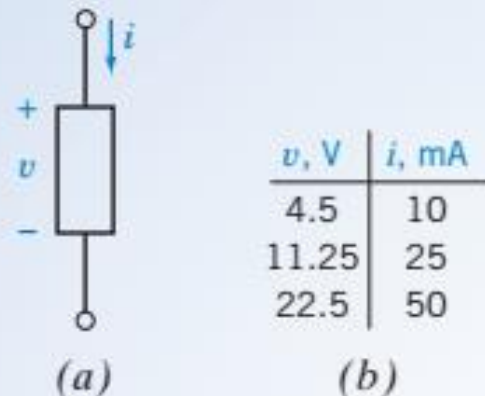
$$v = i^2$$

Determine whether this device is linear.

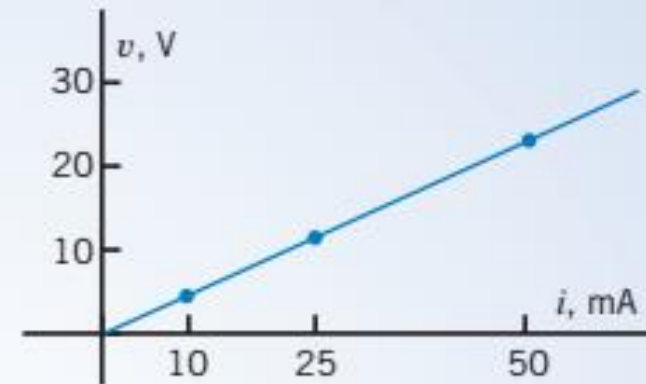


### EXAMPLE 2.2-3 A Model of a Linear Device

A linear element has voltage  $v$  and current  $i$  as shown in Figure 2.2-2a. Values of the current  $i$  and corresponding voltage  $v$  have been tabulated as shown in Figure 2.2-2b. Represent the element by an equation that expresses  $v$  as a function of  $i$ . This equation is a model of the element. Use the model to predict the value of  $v$  corresponding to a current of  $i = 100$  mA and the value of  $i$  corresponding to a voltage of  $v = 18$  V.



**FIGURE 2.2-2** (a) A linear circuit element and (b) a tabulation of corresponding values of its voltage and current.



**FIGURE 2.2-3** A plot of voltage versus current for the linear element from Figure 2.2-2.

### EXAMPLE 2.2-3 A Model of a Linear Device

#### Solution

Figure 2.2-3 is a plot of the voltage  $v$  versus the current  $i$ . The points marked by dots represent corresponding values of  $v$  and  $i$  from the rows of the table in Figure 2.2-2b. Because the circuit element is linear, we expect these points to lie on a straight line, and indeed they do. We can represent the straight line by the equation

$$v = mi + b$$

where  $m$  is the slope and  $b$  is the  $v$ -intercept. Noticing that the straight line passes through the origin,  $v = 0$  when  $i = 0$ , we see that  $b = 0$ . We are left with

$$v = mi$$

The slope  $m$  can be calculated from the data in any two rows of the table in Figure 2.2-2b. For example:

$$\frac{11.25 - 4.5}{25 - 10} = 0.45 \frac{\text{V}}{\text{mA}}, \frac{22.5 - 11.25}{50 - 25} = 0.45 \frac{\text{V}}{\text{mA}}, \text{ and } \frac{22.5 - 4.5}{50 - 10} = 0.45 \frac{\text{V}}{\text{mA}}$$

Consequently,

$$m = 0.45 \frac{\text{V}}{\text{mA}} = 450 \frac{\text{V}}{\text{A}}$$

and

$$v = 450i$$

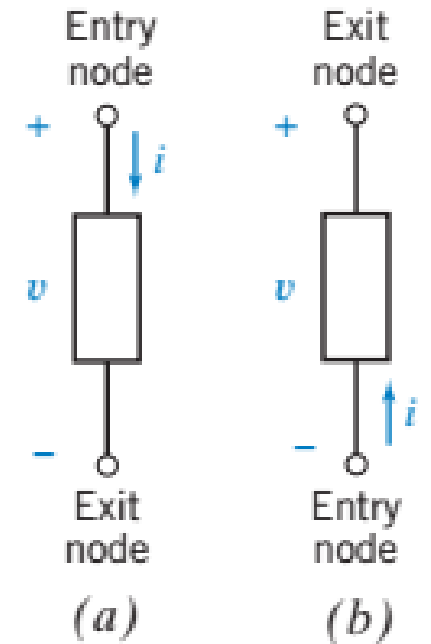
This equation is a model of the linear element. It predicts that the voltage  $v = 450(0.1) = 45 \text{ V}$  corresponds to the current  $i = 100 \text{ mA} = 0.1 \text{ A}$  and that the current  $i = 18/450 = 0.04 \text{ A} = 40 \text{ mA}$  corresponds to the voltage  $v = 18 \text{ V}$ .

## 2.3 Active and Passive Circuit Elements

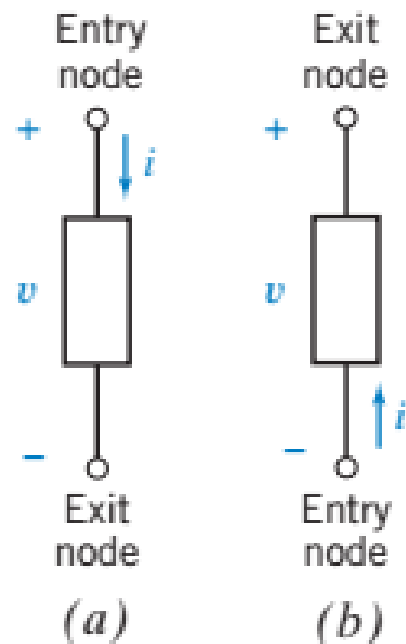
A **passive element** absorbs energy.

An **active element** is capable of supplying energy.

$$w = \int_{-\infty}^t vi \, d\tau \geq 0$$



## 2.3 Active and Passive Circuit Elements



$$w = \int_{-\infty}^t vi \, d\tau \geq 0$$

An element is said to be *active* if it is capable of delivering energy. Thus, an active element violates Eq. 2.3-1 when it is represented by Figure 2.3-1a. In other words, an active element is one that is capable of generating energy. Active elements are potential sources of energy, whereas passive elements are sinks or absorbers of energy. Examples of active elements include batteries and generators. Consider the element shown in Figure 2.3-1b. Note that the current flows into the negative terminal and out of the positive terminal.

**EXAMPLE 2.3-1** An Active Circuit Element

A circuit has an element represented by Figure 2.3-1b where the current is a constant 5 A and the voltage is a constant 6 V. Find the energy supplied over the time interval 0 to  $T$ .

**Solution**

Because the current enters the negative terminal, the energy *supplied* by the element is given by

$$w = \int_0^T (6)(5)d\tau = 30T \text{ J}$$

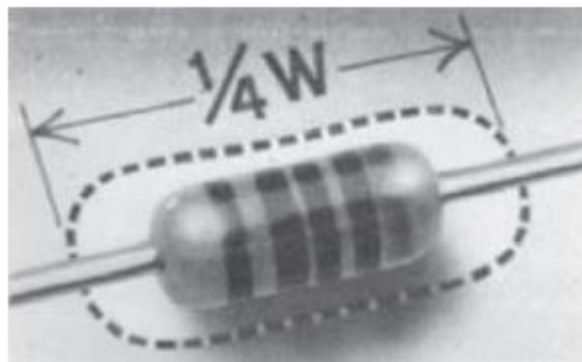
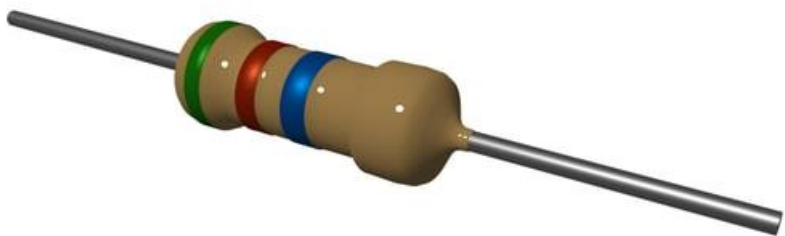
Thus, the device is a generator or an active element, in this case a dc battery.

## 2.4 Resistors

The ability of a material to resist the flow of charge is called its *resistivity*,  $\rho$ . Materials that are good electrical insulators have a high value of resistivity. Materials that are good conductors of electric current have low values of resistivity. Resistivity values for selected materials are given in Table 2.4-1. Copper is commonly used for wires because it permits current to flow relatively unimpeded. Silicon is commonly used to provide resistance in semiconductor electric circuits. Polystyrene is used as an insulator.

**Table 2.4-1 Resistivities of Selected Materials**

| MATERIAL    | RESISTIVITY $\rho$ (OHM.CM) |
|-------------|-----------------------------|
| Polystyrene | $1 \times 10^{18}$          |
| Silicon     | $2.3 \times 10^5$           |
| Carbon      | $4 \times 10^{-3}$          |
| Aluminum    | $2.7 \times 10^{-6}$        |
| Copper      | $1.7 \times 10^{-6}$        |



**Resistance** is the physical property of an element or device that impedes the flow of current; it is represented by the symbol  $R$ .



**FIGURE 2.4-1**

Georg Simon Ohm (1787–1854), who determined Ohm’s law in 1827. The ohm was chosen as the unit of electrical resistance in his honor.

Georg Simon Ohm was able to show that the current in a circuit composed of a battery and a conducting wire of uniform cross-section could be expressed as

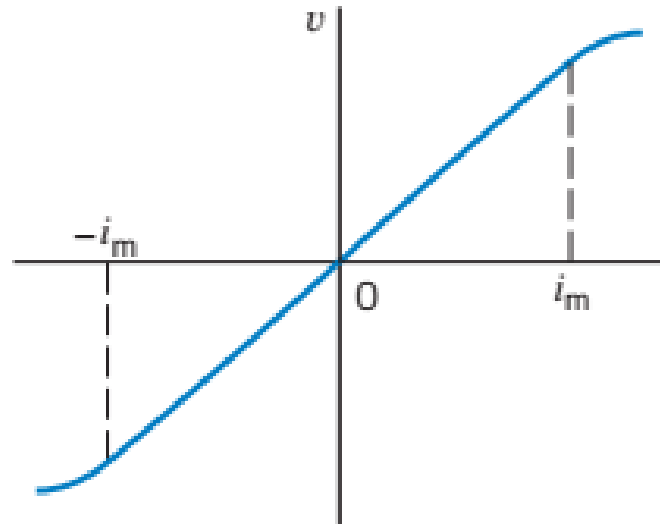
$$i = \frac{Av}{\rho L} \quad (2.4-1)$$

where  $A$  is the cross-sectional area,  $\rho$  the resistivity,  $L$  the length, and  $v$  the voltage across the wire element. Ohm, who is shown in Figure 2.4-1, defined the constant resistance  $R$  as

$$R = \frac{\rho L}{A} \quad (2.4-2)$$

Ohm’s law, which related the voltage and current, was published in 1827 as

$$v = Ri \quad (2.4-3)$$



**FIGURE 2.4-3** A resistor operating within its specified current range,  $\pm i_m$ , can be modeled by Ohm's law.



**FIGURE 2.4-4** A resistor with element current and element voltage.

$$i_a = -i_b$$

The element current  $i_a$  and the element voltage  $v$  adhere to the passive convention,

$$v = Ri_a$$

Replacing  $i_a$  by  $-i_b$  gives

$$v = -Ri_b$$

Ohm's law, Eq. 2.4-3, can also be written as

$$i = Gv \quad (2.4-5)$$

where  $G$  denotes the *conductance* in siemens (S) and is the reciprocal of  $R$ ; that is,  $G = 1/R$ . Many engineers denote the units of conductance as mhos with the  $\mathcal{O}$  symbol, which is an inverted omega (mho is *ohm* spelled backward). However, we will use SI units and retain siemens as the units for conductance.

The power delivered to a resistor (when the passive convention is used) is

$$p = vi = v\left(\frac{v}{R}\right) = \frac{v^2}{R} \quad (2.4-6)$$

Alternatively, because  $v = iR$ , we can write the equation for power as

$$p = vi = (iR)i = i^2R \quad (2.4-7)$$

Thus, the power is expressed as a nonlinear function of the current  $i$  through the resistor or of the voltage  $v$  across it.

### EXAMPLE 2.4-1 Power Dissipated by a Resistor

Let us devise a model for a car battery when the lights are left on and the engine is off. We have all experienced or seen a car parked with its lights on. If we leave the car for a period, the battery will run down or go dead. An auto battery is a 12-V constant-voltage source, and the lightbulb can be modeled by a resistor of 6 ohms. The circuit is shown in Figure 2.4-8. Let us find the current  $i$ , the power  $p$ , and the energy supplied by the battery for a four-hour period.

#### Solution

According to Ohm’s law, Eq. 2.4-3, we have

$$v = Ri$$

Because  $v = 12 \text{ V}$  and  $R = 6 \Omega$ , we have  $i = 2 \text{ A}$ .

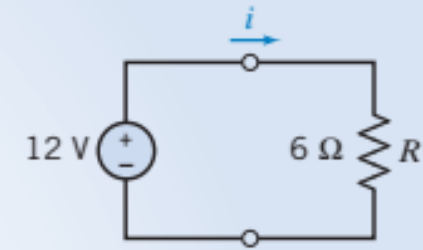
To find the power delivered by the battery, we use

$$p = vi = 12(2) = 24 \text{ W}$$

Finally, the energy delivered in the four-hour period is

$$w = \int_0^t p d\tau = 24t = 24(60 \times 60 \times 4) = 3.46 \times 10^5 \text{ J}$$

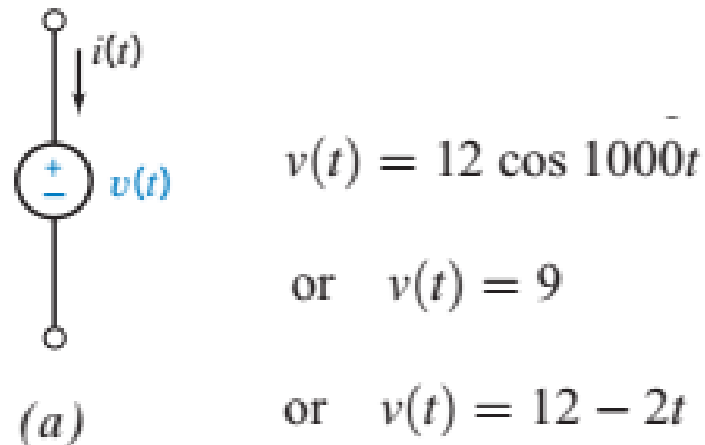
Because the battery has a finite amount of stored energy, it will deliver this energy and eventually be unable to deliver further energy without recharging. We then say the battery is run down or dead until recharged. A typical auto battery may store  $10^6 \text{ J}$  in a fully charged condition.



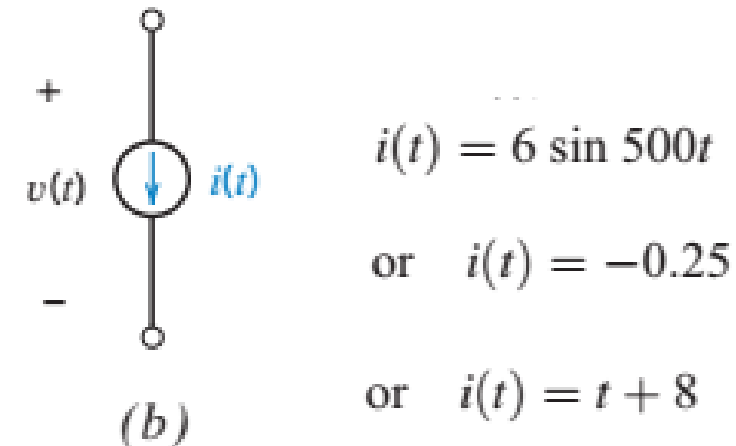
**FIGURE 2.4-8** Model of a car battery and the headlight lamp.

## 2.5 Independent Sources

Some devices are intended to supply energy to a circuit. These devices are called sources. Sources are categorized as being one of two types: voltage sources and current sources. Figure 2.5-1a shows the symbol that is used to represent a voltage source. The voltage of a voltage source is specified, but the current is determined by the rest of the circuit. A voltage source is described by specifying the function  $v(t)$ , for example,

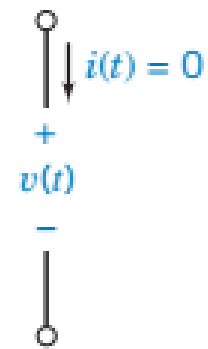


(a) Voltage source.

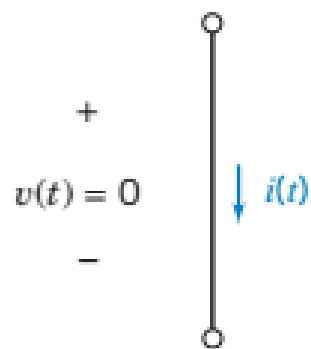


(b) Current source.

An **independent source** is a voltage or current generator not dependent on other circuit variables.



(a)



(b)

The short circuit and open circuit are special cases of ideal sources. A *short circuit* is an ideal voltage source having  $v(t) = 0$ . The current in a short circuit is determined by the rest of the circuit. An *open circuit* is an ideal current source having  $i(t) = 0$ . The voltage across an open circuit is determined by the rest of the circuit. Figure 2.5-3 shows the symbols used to represent the short circuit and the open circuit. Notice that the power absorbed by each of these devices is zero.

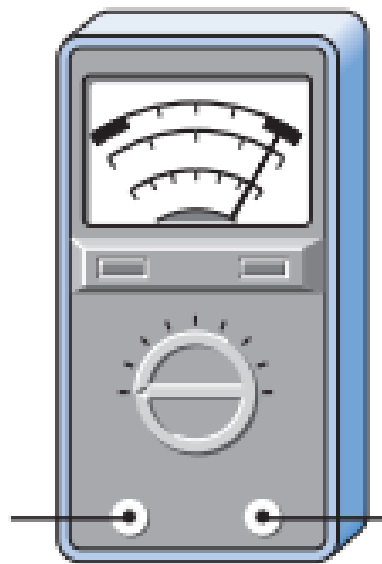
Open and short circuits can be added to a circuit without disturbing the branch currents and voltages of all the other devices in the circuit. Figure 2.6-3 shows how this can be done. Figure 2.6-3a shows an example circuit. In Figure 2.6-3b an open circuit and a short circuit have been added to this example circuit. The open circuit was connected between two nodes of the original circuit. In contrast, the short circuit was added by cutting a wire and inserting the short circuit. Adding open circuits and short circuits to a network in this way does not change the network.

Open circuits and short circuits can also be described as special cases of resistors. A resistor with resistance  $R = 0$  ( $G = \infty$ ) is a short circuit. A resistor with conductance  $G = 0$  ( $R = \infty$ ) is an open circuit.

**FIGURE 2.5-3**

(a) Open circuit.

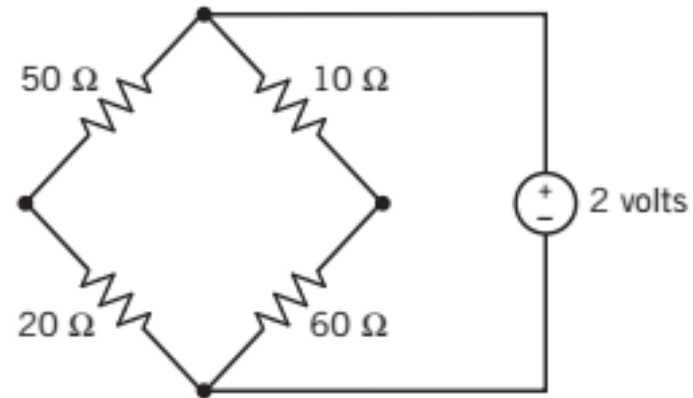
(b) Short circuit.



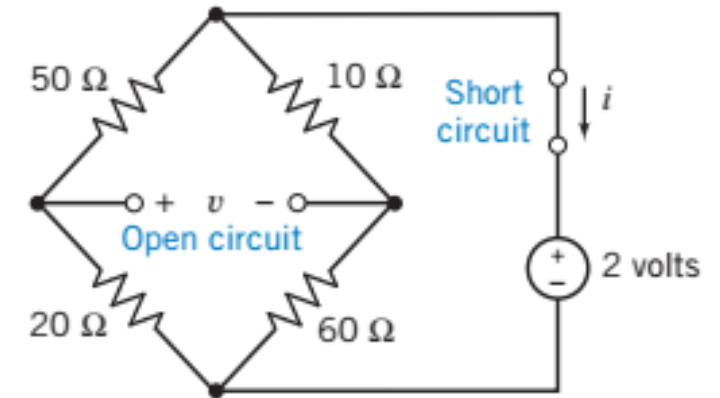
(a)



(b)

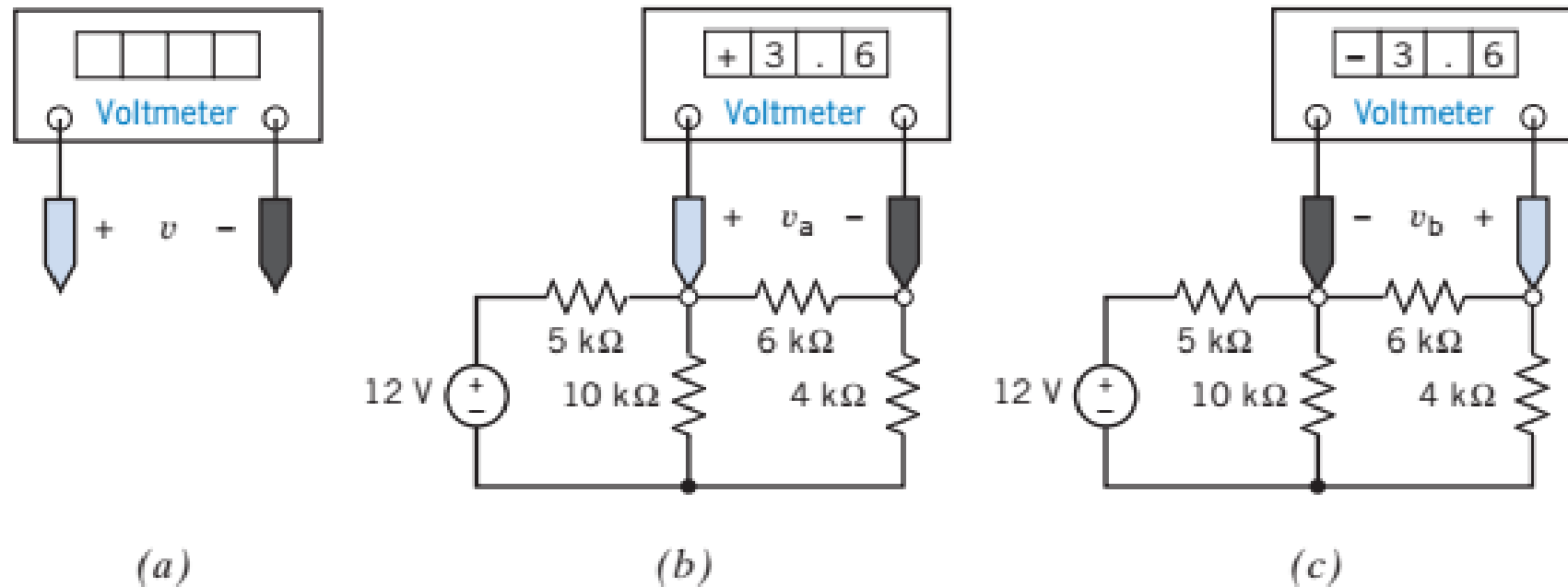


(a)



(b)

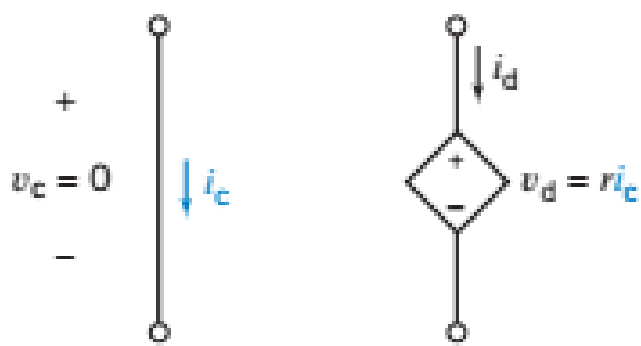
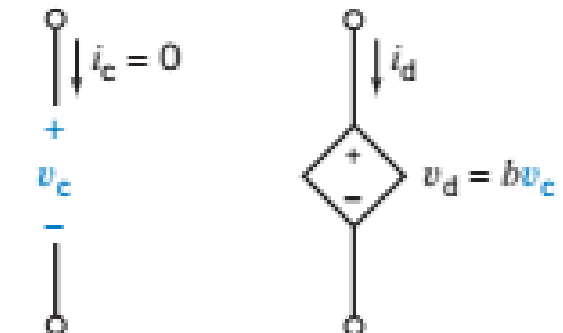
**FIGURE 2.6-1** (a) A direct-reading (analog) meter.  
(b) A digital meter.



**FIGURE 2.6-4** (a) The correspondence between the color-coded probes of the voltmeter and the reference direction of the measured voltage. In (b), the + sign of  $v_a$  is on the left, whereas in (c), the + sign of  $v_b$  is on the right. The colored probe is shown here in blue. In the laboratory this probe will be red. We will refer to the colored probe as the “red probe.”

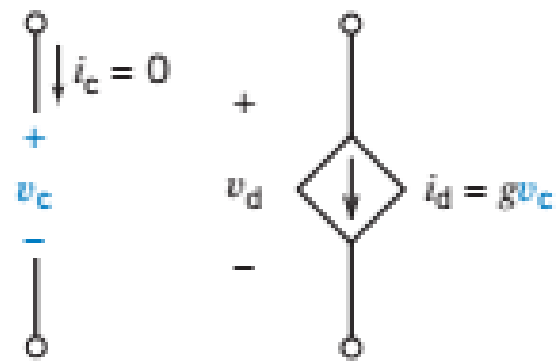
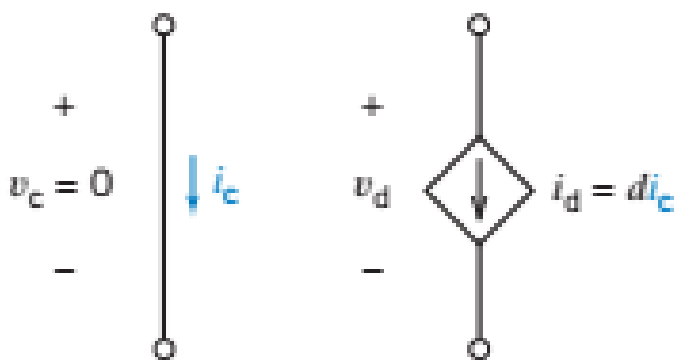
## 2.7 Dependent Sources

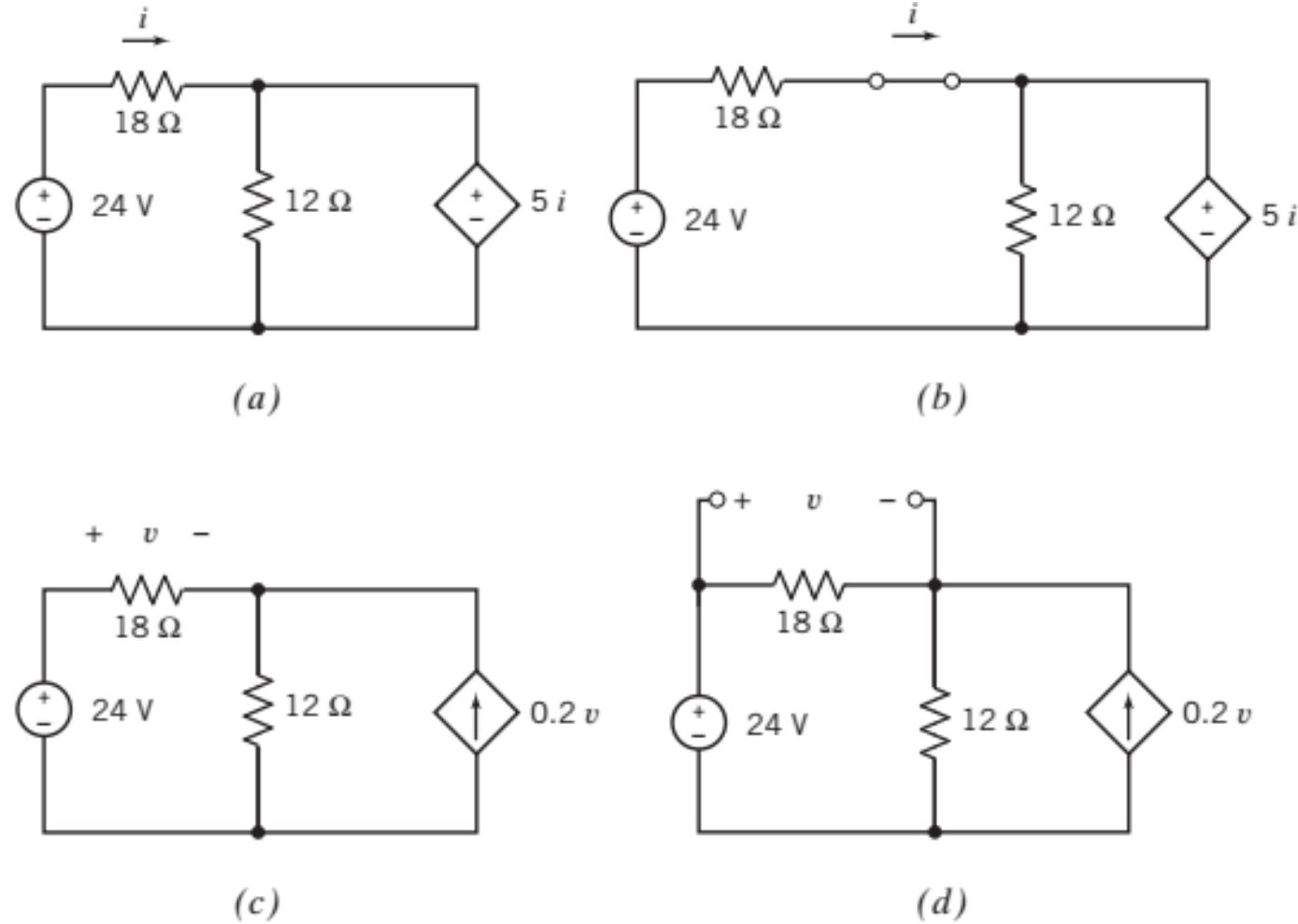
**Table 2.7-1** Dependent Sources

| DESCRIPTION                                                                                                                                     | SYMBOL                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <p>Current-Controlled Voltage Source (CCVS)</p> <p><math>r</math> is the gain of the CCVS.</p> <p><math>r</math> has units of volts/ampere.</p> |   |
| <p>Voltage-Controlled Voltage Source (VCVS)</p> <p><math>b</math> is the gain of the VCVS.</p> <p><math>b</math> has units of volts/volt.</p>   |  |

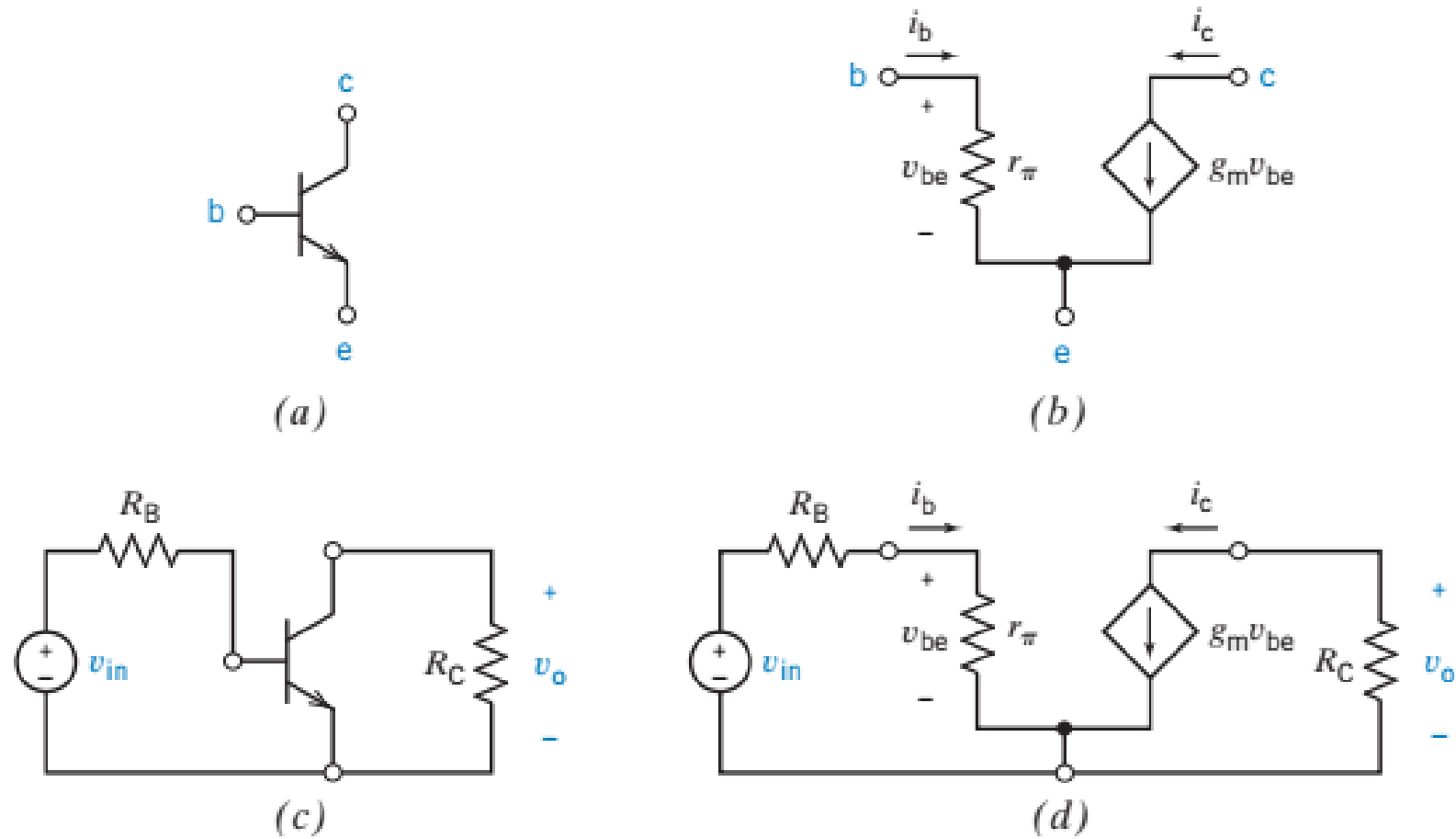
## 2.7 Dependent Sources

**Table 2.7-1** Dependent Sources

| DESCRIPTION                                                                                                                                       | SYMBOL                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <p>Voltage-Controlled Current Source (VCCS)</p> <p><math>g</math> is the gain of the VCCS.</p> <p><math>g</math> has units of amperes/volt.</p>   |   |
| <p>Current-Controlled Current Source (CCCS)</p> <p><math>d</math> is the gain of the CCCS.</p> <p><math>d</math> has units of amperes/ampere.</p> |  |



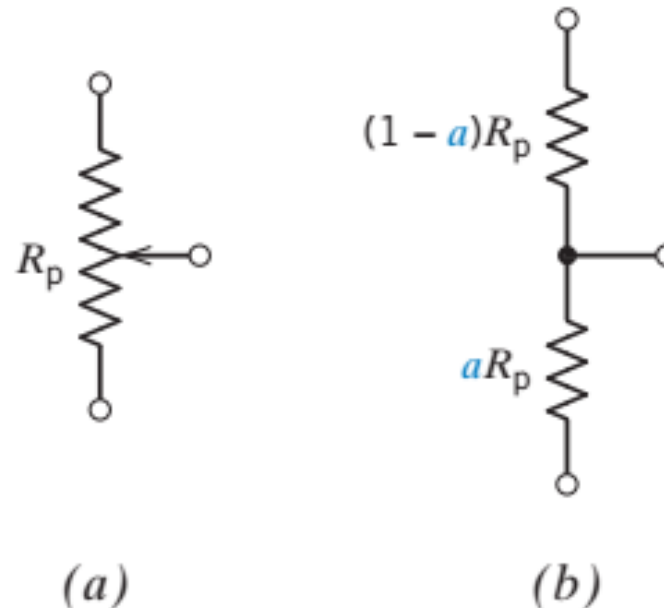
**FIGURE 2.7-1** The controlling current of a dependent source shown as (a) the current in an element and as (b) the current in a short circuit in series with that element. The controlling voltage of a dependent source shown as (c) the voltage across an element and as (d) the voltage across an open circuit in parallel with that element.



**FIGURE 2.7-2** (a) A symbol for a transistor. (b) A model of the transistor. (c) A transistor amplifier. (d) A model of the transistor amplifier.

## 2.8 *Transducers*

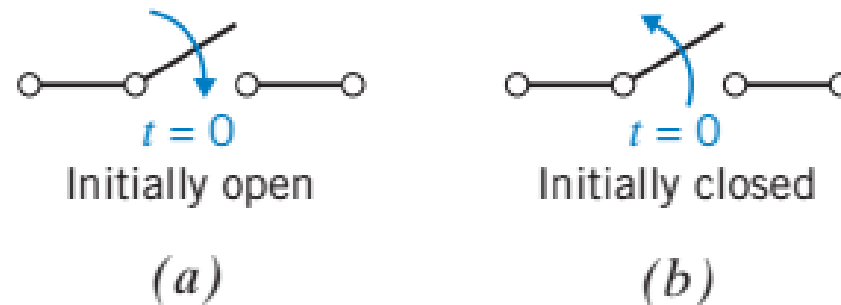
Transducers are devices that convert physical quantities to electrical quantities. This section describes two transducers: potentiometers and temperature sensors. Potentiometers convert position to resistance, and temperature sensors convert temperature to current.



**FIGURE 2.8-1** (a) The symbol and (b) a model for the potentiometer.

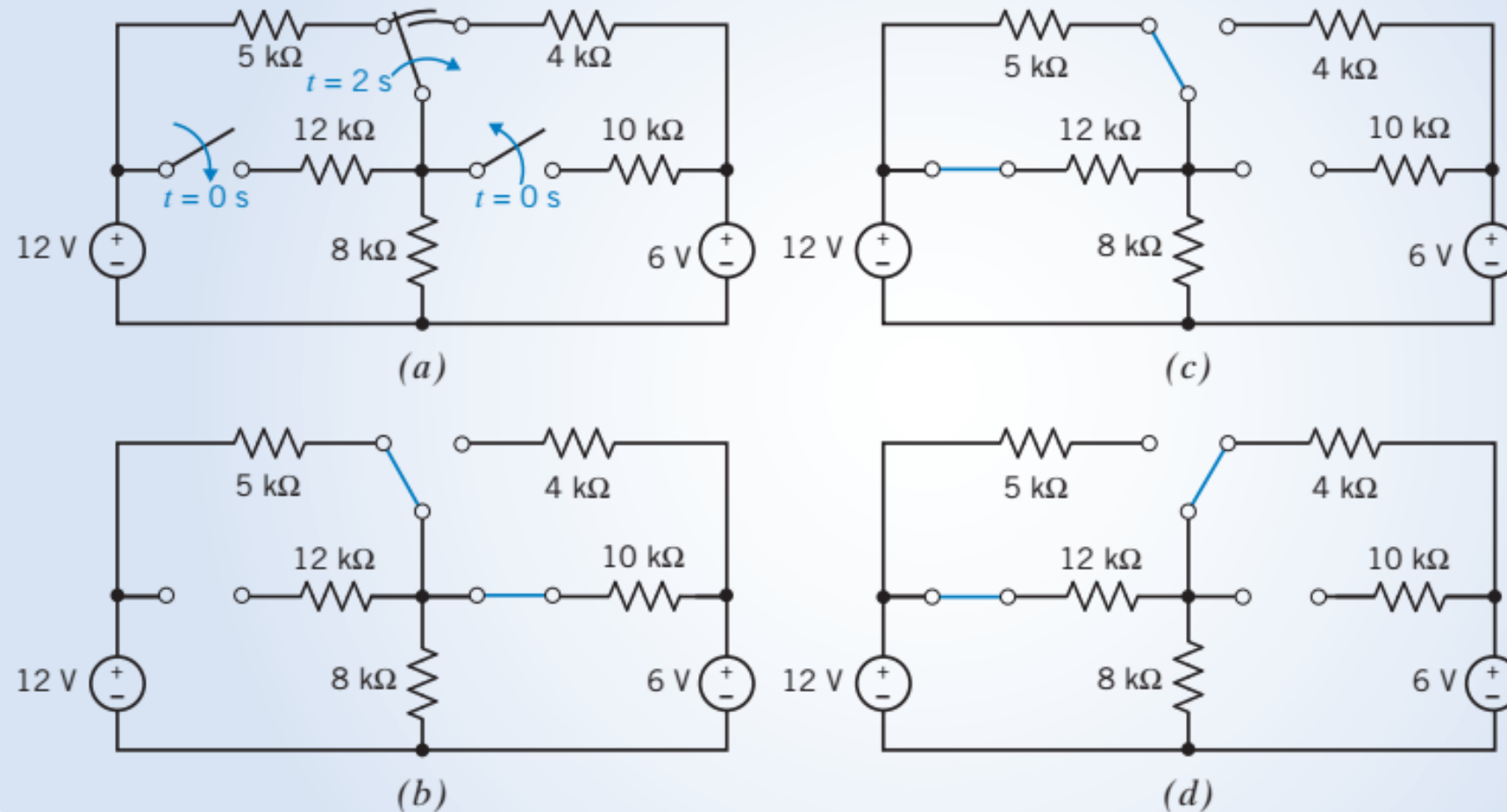
## 2.9 Switches

Switches have two distinct states: open and closed. Ideally, a switch acts as a short circuit when it is closed and as an open circuit when it is open.



**FIGURE 2.9-1** SPST switches. (a) Initially open and (b) initially closed.

## 2.9 Switches



**FIGURE 2.9-3** (a) A circuit containing several switches. (b) The equivalent circuit for  $t \leq 0$  s. (c) The equivalent circuit for  $0 < t < 2$  s. (d) The equivalent circuit for  $t > 2$  s.